System Design Specification

for

Employee Performance Review (EPR) Dashboard and Workflow System

Version 0.2 Approve Pending

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Document Control

Version	Date	Author	Summary
1.0	10/16/2025	Momoka Aung, Chiemela Eziechile	Initial draft.
2.0	10/20/2025	Hoang Le (Dempsey)	Updated 3.2.3 (HR/EPR pages); added KPI definitions and why they matter; UX notes; added flow caption.
3.0	12/01/2025	Momoka Aung	Added detailed information of System Strategy and Architecture part.
4.0	12/03/2025	Momoka Aung	Updated the documents with work flow diagrams and automation flow diagrrams.
5.0	12/03/2025	Momoka Aung	Update Box Upload function to the current version.
6.0	03/23/2026	Momoka Aung	Added Section 9 Database Design.
7.0	04/20/2026	Hoang Le (Dempsey)	Updated Section 9 User Interface with expanded Power BI dashboard documentation for Personnel Matters and Vacancies & Recruitment, including Smartsheet data, intake forms, tracker screenshots, and dashboard page descriptions.

1. Introduction

1.1. Purpose

The purpose of this document is to define the technical architecture and detailed design of the Intelligent HR Analytics Platform developed for the Santa Barbara County Public Defender's Office (SBCPDO). This Software Design Specification (SDS) translates the approved Software Requirements Specification (SRS) into a concrete system design that guides system implementation, integration, testing, and maintenance.

1.2. Scope of Coverage

This SDS describes the design of the platform's core components, including workflow automation for Employee Performance Reviews (EPR), Personnel Matters, Vacancies and Recruitments, and Employee Separations. It also outlines document management integration with Box, reporting through Power BI, and role-based access controls. Business policies and operational procedures are defined in the SRS and are not repeated here.

1.3. Document Conventions

This document follows a hierarchical numbering structure (e.g., 1.0, 1.1, 1.2). Formatting conventions are used to distinguish elements of the design:

- **Bold text** indicates section titles or key architectural components.
- *Italic text* refers to external documents such as the SRS.
- `Monospaced text` represents field names, configuration values, or identifiers.

1.4. Intended Audience

This document is intended for developers, system architects, testers, and technical staff responsible for implementing or maintaining the system. Project managers and administrators may also reference this document to understand the system's architecture. Non-technical stakeholders should refer to the SRS for functional and business-level descriptions.

1.5. System Overview

The Intelligent HR Analytics Platform is a web-based system that automates and centralizes key HR workflows within SBCPDO. The system supports performance reviews, personnel case management, recruitment tracking, and employee separations through integrated workflows and reporting dashboards.

The architecture consists of several logical layers: a workflow layer for HR processes, an integration layer connecting external services, a data layer for structured records and document links, an analytics layer for reporting, and a security layer enforcing role-based access control. Together, these components provide a scalable and maintainable platform for managing HR operations and generating leadership insights.

2. Design Considerations

This section describes the key assumptions, constraints, and design goals that influence the architecture and implementation of the Intelligent HR Analytics Platform.

2.1. Assumptions and Dependencies

Technologies and platforms used in the system include:

- **Smartsheet**: Serves as the primary workflow engine for tracking EPRs, personnel matters, vacancies and recruitments, and employee separations.
- **Box**: Used as the secure document repository for finalized HR documents and records.
- **Power BI**: Provides reporting dashboards and analytics for HR leadership.
- **Email/Calendar Systems**: Send automated notifications and reminders.
- **AWS Lambda**: Executes serverless functions to automate backend processes and event-driven tasks.

Additional assumptions include:

- All services operate within cloud-based environments and require active enterprise licenses.
- Users access the system through modern web browsers with internet connectivity.
- HR maintains accurate employee and supervisor information for proper workflow routing.

2.2. General Constraints

- System availability depends on Smartsheet, Box, and Power BI service uptime.
- API rate limits and vendor service restrictions may affect automation performance.
- Access to the system is limited to authorized County staff through Single Sign-On (SSO).
- Customization is limited to the capabilities provided by the supported enterprise platforms.

2.3. Goals and Guidelines

- Centralize HR workflow tracking for performance reviews, personnel matters, recruitment activities, and employee separations.
- Improve visibility and accountability across HR processes.
- Provide automated reminders and status tracking to ensure timely completion of tasks.
- Maintain secure document storage and reporting capabilities aligned with County policies.

2.4. Development Methods

An Agile development approach was followed. System features were implemented and tested through iterative development cycles, allowing continuous feedback from project sponsors and HR staff. Responsibilities were divided among team members to support parallel development and faster integration of system components.

3. Architectural Strategies

This section describes the major design strategies used in the Intelligent HR Analytics Platform, including technology selection, system reuse, integration methods, and scalability considerations.

3.1. Technology Stack Selection

- The system is built using cloud-based enterprise platforms including Smartsheet, Box, and Power BI.
- Smartsheet serves as the workflow engine for tracking Employee Performance Reviews (EPR), Personnel Matters, Vacancies and Recruitment activities, and Employee Separations.
- Box provides secure document storage and archiving for finalized HR records.
- Power BI generates dashboards and analytics reports for HR leadership and management.
- AWS Lambda supports serverless, event-driven automation and system integrations across the platform.
- These platforms were selected because they integrate with existing County infrastructure and allow workflow automation without extensive custom development.

3.2. Reuse of Existing Components

- The platform leverages existing enterprise tools already approved by the County rather than building new infrastructure from scratch.
- Smartsheet automation rules are reused across multiple workflows to standardize tracking and notifications.
- Existing Box storage policies are used to manage document retention and access control.

3.3. Integration and Communication

- Smartsheet integrates with Box to store finalized documents and maintain secure document links.
- Power BI connects to Smartsheet datasets to generate reports and dashboards.
- AWS Lambda facilitates event-driven integrations and automation between systems.
- Automated notifications and reminders are delivered through email.
- All integrations occur through secure APIs using HTTPS and OAuth authentication.

3.4. Modularity and Scalability

- The system follows a modular design where each workflow (EPR, Personnel Matters, Vacancies and Recruitment, and Separations) operates as an independent subsystem.
- Each subsystem maintains its own tracker sheets and automation rules while sharing the same architecture.
- Additional HR workflows can be added without major structural changes.

3.5. Data Storage and Management

- Smartsheet stores operational workflow data and status tracking information.
- Box stores finalized documents and case files.
- Power BI consumes structured datasets to provide reporting and analytics.

3.6. Security Strategy

- Role-Based Access Control (RBAC) restricts system access based on user role.
- All integrations use encrypted HTTPS connections and OAuth authentication.
- Audit logging is enabled to track workflow updates and document access.

3.7. Rejected Alternatives

The following architectural approaches were evaluated but not implemented:

- Manual PDF routing via email: lacked centralized tracking and analytics capability.
- DocuSign for Digital Signatures: Evaluated as a potential solution for document signing; however, it was not adopted due to certification and compliance limitations within the County environment.

4. System Architecture

The Intelligent HR Analytics Platform is implemented using a modular, SaaS-based distributed architecture. The system is decomposed into interoperable subsystems that communicate through authenticated API integrations and native platform connectors.

The architecture supports multiple HR workflow domains while maintaining a shared integration and security foundation.

4.1. System Decomposition

The system is divided into functional subsystems built on a shared integration framework.

4.1.1. Workflow Subsystems (Smartsheet-Based)

Each HR initiative is implemented as an independent workflow module within Smartsheet. All workflow modules share a consistent structural pattern:

- Dedicated tracking sheet(s)
- Structured column schema
- State-controlled `Status` field
- Automation rule groups
- Approval routing logic
- Archival trigger

EPR Workflow Subsystem Manages review creation, supervisor assignment, approval routing, digital acknowledgments, tracking completion status, and initiating archival.

Personnel Matters Subsystem Handles administrative record tracking, document management, optional metadata extraction, approval routing, and archival.

Vacancies and Recruitment Subsystem Creates and tracks recruitment records, manages hiring status and key steps.

Separations Subsystem Implements structured separation intake, validation logic, outbound communication tracking, status management, and archival initiation.

4.1.2. Secure Document Repository (Box.com)

Serves as the centralized storage layer for finalized HR documents. Responsibilities include:

- Structured folder hierarchy enforcement
- Role-based access control
- Retention policy enforcement
- Audit logging

4.1.3. Analytics and Reporting Subsystem (Power BI)

Provides cross-workflow dashboards and reporting capabilities. Responsibilities include:

- Dataset ingestion from Smartsheet
- Data modeling and normalization
- KPI computation
- Interactive dashboard presentation

4.1.4. Integration and Data Services Layer

Coordinates secure data flow between Smartsheet, Box.com, external APIs, and Power BI using REST APIs, OAuth-based authentication, and scheduled synchronization mechanisms.

4.2. Component Responsibilities

Each architectural component maintains defined boundaries and roles.

4.2.1. Smartsheet

- Operational workflow engine
- Primary system of record for structured HR metadata
- Automation rule execution
- Approval routing and status transitions
- Notification generation

4.2.2. Box.com

- Immutable document repository
- Structured folder organization
- Metadata tagging
- Access control enforcement
- Activity logging

4.2.3. Power BI

- Analytical data modeling
- KPI and metric computation
- Dashboard visualization
- Scheduled dataset refresh

4.2.4. Integration Layer

- API invocation and token handling
- Data validation and transformation (if required)
- Retry handling for transient failures
- Synchronization status logging

4.3. Communication and Control Flow

All workflow domains follow a standardized event-driven lifecycle model.

4.3.1. Workflow Lifecycle

1. **Initiation** — Triggered by form submission, row creation, or time-based automation.
2. **Processing** — Automation rules evaluate conditions and update state.
3. **Approval Routing** — Sequential or single-stage approvals executed where applicable.
4. **Completion Validation** — Required fields and approvals verified.
5. **Archival Trigger** — Finalized artifacts uploaded to Box; metadata written back.
6. **Analytics Refresh** — Power BI incorporates updated data during scheduled refresh.

Each subsystem adheres to this control model to ensure architectural consistency.

4.4. Data Flow and Integration Logic

The system primarily uses one-way data synchronization patterns to preserve data integrity.

4.4.1. Smartsheet → Box (Archival Flow)

- Trigger: Workflow reaches final completion state.
- Action: Upload finalized document with metadata.
- Write-back: Store Box file URL in Smartsheet.
- Direction: One-way.

4.4.2. Smartsheet → Power BI (Analytics Flow)

- Trigger: Scheduled dataset refresh.
- Action: Import structured workflow data.
- Direction: One-way read-only consumption.

4.4.3. External APIs → Smartsheet (Optional Metadata Extraction)

- Trigger: Document upload event or row update.
- Action: Extract structured fields from document.
- Write-back: Populate normalized Smartsheet columns.

4.4.4. External APIs → Separations (Email Notifications)

- Trigger: Separation request initiated or status updated.
- Action: Generate and send customized email notifications.
- Data Source: Smartsheet records and associated documents (Box).
- Delivery: Email system via secure API.

All integrations use HTTPS encryption and OAuth-based authentication.

4.5. Scalability and Modularity

4.5.1. Subsystem Isolation

Workflow domains operate independently using separate sheets and automation rule groups. Changes to one subsystem do not require structural changes to others.

4.5.2. Scalability Approach

The architecture supports scaling through:

- Increased row volume within Smartsheet trackers
- Expanded Box folder hierarchies
- Enlarged Power BI datasets

No architectural redesign is required for projected growth within expected operational bounds.

4.6. Justification of Architecture

The selected architecture satisfies the following objectives:

- **Scalability** — Configuration-based growth without backend expansion.
- **Reusability** — Shared integration and workflow patterns across domains.
- **Security** — OAuth-based authentication and role-based access control.
- **Maintainability** — Low-code automation reduces long-term technical debt.
- **Auditability** — Structured tracking and logging across all workflows.

5. Design Policies and Implementation Tactics

This section defines the technical standards, architectural policies, and implementation decisions governing the Intelligent HR Analytics Platform. These guidelines ensure consistency, maintainability, and security across all workflow domains.

5.1. Platform Components

5.1.1. Smartsheet (Workflow Engine)

Smartsheet functions as the operational workflow engine and system of record for structured HR workflow data. It manages workflow state, routing logic, and automation rules through controlled `Status` fields. Only structured metadata is stored within Smartsheet. Finalized document artifacts are stored externally, with references (e.g., `Box_URL`) maintained in workflow records.

5.1.2. Box.com (Document Repository)

Box provides secure storage for finalized workflow documents such as performance reviews, personnel files, recruitment artifacts, and separation documentation. Folder-level permissions and metadata tagging support controlled access, document retrieval, and retention policy enforcement.

5.1.3. Power BI (Analytics Engine)

Power BI provides analytics and reporting capabilities. Operational data from Smartsheet is modeled and transformed within Power BI to generate dashboards and performance metrics using DAX measures. Power BI operates in read-only mode with respect to workflow data.

5.1.4. Integration and Authentication

System integrations use secure HTTPS communication with OAuth 2.0 authentication. Access permissions follow the principle of least privilege, and credentials are never embedded directly in configuration files or scripts.

5.2. Requirements Traceability

Traceability between requirements and implementation is maintained through a Requirements Traceability Matrix (RTM). Each requirement maps to one or more implementation artifacts, including Smartsheet columns, automation rules, Box metadata fields, or Power BI report elements. Traceability is verified prior to major system releases.

5.3. Verification and Validation Strategy

System validation is performed at multiple levels to ensure functional correctness and data reliability.

5.3.1. Unit-Level Validation

Individual configuration elements are tested using sample data. Automation rules, column formulas, and Power BI measures are verified to ensure expected behavior and accurate calculations.

5.3.2. Testing

- **Unit Testing:** Individual components and workflows were tested to ensure correct functionality (e.g., form submissions, data updates, and automation logic).
- **Integration Testing:** System integrations between Smartsheet, Box, Power BI, and external APIs were tested to ensure proper data flow and communication.
- **Performance Testing:** Basic validation was conducted to ensure the system performs reliably under expected usage conditions.

5.4. Error Handling and Exception Management

- API failures are logged and retried where applicable.
- Data validation errors are flagged within Smartsheet workflows.
- Notification failures (e.g., email delivery issues) are monitored and logged.

5.5. Logging and Monitoring

- System activities and API interactions are logged for auditing and troubleshooting.
- Workflow execution and automation processes are monitored for failures.
- Logs are reviewed periodically to ensure system reliability.

5.6. Backup and Recovery

- Data stored in Box and Smartsheet relies on platform-managed backups.
- Critical records can be restored based on platform retention policies.
- Periodic data exports may be used for additional backup assurance.

5.7. Access Control

- Role-based access control is supported across Smartsheet, Box, and Power BI.
- User roles and permissions (e.g., HR staff, supervisors, administrators) will be configured based on organizational policies.
- Final access control settings will be defined and adjusted during system handoff to the sponsoring organization to align with internal security requirements.

5.7.1. Accessibility Validation

Dashboards and user interfaces are evaluated for accessibility, including appropriate color contrast, keyboard navigation support, screen-reader compatibility, and accessible export formats.

5.7.2. Data Integrity Monitoring

Data consistency between Smartsheet and Power BI is monitored through periodic comparisons of row counts and aggregate metrics. Significant discrepancies trigger manual review.

5.8. Engineering Trade-Off Decisions

5.8.1. Low-Code vs. Custom Development

The system prioritizes low-code platform capabilities to reduce infrastructure complexity and simplify long-term maintenance. Custom scripting is used only where required for integration or specialized validation.

5.8.2. Analytics Refresh Strategy

Analytics data is refreshed on a scheduled basis rather than in real time. Outside of work hours Power BI refresh cycles provide sufficient reporting accuracy while reducing API usage and operational overhead.

5.8.3. Source-of-Truth Model

The system maintains clear ownership of data across subsystems:

- Smartsheet — operational workflow data
- Box.com — finalized document repository
- Power BI — analytics and reporting

This separation prevents conflicting updates and maintains clear system responsibilities.

5.9. Design Standards

5.9.1. Date Handling

All system dates follow the ISO 8601 standard in UTC. Localization occurs only within presentation layers such as Power BI dashboards.

5.9.2. Box Metadata Policy

Archived documents must include metadata identifying the associated workflow record (e.g., EmployeeID or VacancyID), workflow domain, and retention classification. These metadata fields support auditability and retrieval.

5.10. Communication Protocols

All system communications occur over HTTPS using TLS 1.2 or higher. Integrations use OAuth 2.0 authentication and vendor-supported APIs or connectors. Audit logging is maintained according to county policy.

5.11. Architectural Patterns

The system architecture follows several design principles:

- **Event-driven workflows** triggered by status changes, dates, or form submissions.
- **Idempotent processing** to prevent duplicate actions or artifacts.
- **Retry mechanisms** for handling transient integration failures.
- **Separation of concerns** between workflow management, document storage, and analytics.

5.12. Abstraction Strategy

External systems are accessed through APIs and platform connectors, allowing workflow logic to remain independent of vendor-specific implementations. In Power BI, reusable DAX measures encapsulate business logic, simplifying report maintenance and ensuring consistency across dashboards.

6. Detailed System Design

All workflow domains follow a consistent structural pattern. Each workflow is implemented as a Smartsheet-based tracker with a controlled lifecycle managed through the `Status` field. Automation rules handle routing, notifications, and validation, while completed records trigger document archival in Box and reporting updates in Power BI.

- Structured Smartsheet tracker
- State-based workflow management via `Status`
- Automation rule groups for routing and notifications
- Archival trigger to Box
- Reporting integration with Power BI

6.1. EPR Workflow

Purpose Manages the full lifecycle of Employee Performance Reviews, including creation, routing through supervisory approvals, archival of finalized documents, and reset for future review cycles.

Core Data Fields `EmployeeNumber, FirstName, LastName, JobClass, StartDate, EmploymentStatus, ProbationDueDate, SignedEPRDueDate, PreviousEPRSigned, PreviousEPR HR, Supervisor1-3, HeadHR, HeadOffice, Status, Box_URL`

Workflow States

`Not Created` → `With HR` → `With Supervisor1` → `With Supervisor2` → `With Supervisor3` → `With Head of HR` → `With Head of Office` → `Saving to Box` → `Completed`

Integration Points

- Python-based automation for archival validation
- Box for secure document upload and storage
- Power BI for dataset refresh and reporting analytics

6.2. Personnel Matters Workflow

Purpose Tracks personnel-related administrative cases that require documentation, review, and archival.

Core Data Fields `MatterID, OpenDate, Status, Complaint, Responder, IncidentSummary, IssueType, PolicyReference, Supervisor, NextStep, NextStepDueDate, Box_URL`

Workflow States

`Opened` → `In Progress` → `Closed`

Integration Points

- Optional document metadata extraction via Python-based automation
- Archival to Box repository
- Power BI reporting integration

6.3. Vacancies and Recruitment Workflow

Purpose Tracks recruitment requests and hiring progress from vacancy creation to final position fill.

Core Data Fields `VacancyID, JobClassTitle, Department, Label, VacancyStartDate, InterviewsConducted`

Workflow States

`Posted` → `Screening` → `Interviewing`
→ `Offer Extended` → `Offer Accepted` → `Filled` → `OnHold`

Integration Points

- Optional document extraction via Box webhook and Python-based automation
- Power BI analytics and recruitment metrics

6.4. Separations Workflow

Purpose Tracks employee separation cases and manages communication workflows related to employee departures.

Core Data Fields `EmployeeID, FirstName, LastName, Email, ExitReason, EffectiveDate, LastDayDate, EmailStatus`

Workflow States

`Missing Field` → `Awaiting Email` → `Email Sent` → `Email Failed`

Integration Points

- Automated email generation and delivery
- Optional document processing and extraction
- Power BI reporting integration

7. Execution and Automation Model

This section defines the automation execution logic and shared processing framework.

7.1. Workflow Automation Structure

All workflows implement:

- Event-driven rule execution
- Status-based state transitions
- Validation before archival
- Idempotent integration triggers

Automation rules are organized into logical groups:

- Initialization Rules
- Routing Rules
- Status Management Rules
- Archival Rules
- Reminder/Escalation Rules

Detailed rule configurations are documented in Appendix A.

7.2. Shared Python Execution Framework

The Python execution framework extends native automation capabilities for:

- Box archival synchronization
- Metadata extraction
- Record validation
- Email generation support
- Bulk processing tasks

7.2.1. Execution Pattern

1. Authenticate with external APIs
2. Retrieve eligible records
3. Validate required fields
4. Execute workflow-specific logic
5. Update Smartsheet and/or Box
6. Log results

7.2.2. Security Controls

- OAuth 2.0 authentication
- Environment-based secret storage
- Least-privilege API scopes
- Execution logging for auditability

7.3. Performance Characteristics

- Event-driven, non-blocking execution
- Idempotent processing to prevent duplication
- Linear scaling within API rate limits

8. Database Design

This section describes the structure, purpose, and relationships of the primary Smartsheet data repository used by the Intelligent HR Analytics & EPR Management Dashboard. A single Smartsheet table (Employees) stores employee details, job classifications, and Employee Performance Review (EPR) workflow status. This centralized design simplifies automation, reporting, and integration with Box, and Power BI.

8.1. Data Integration

- Integrated with Power BI for reporting.
- Box stores completed EPR documents.
- Power Automate manages data synchronization across systems.

8.2. Security

- Role-based access (HR Admin, Supervisor, Employee).
- All changes are tracked for audit purposes.
- Secure API access via OAuth 2.0.

8.3. Data Tables

8.3.1. EPR Tracker - Employees Performance Tracker

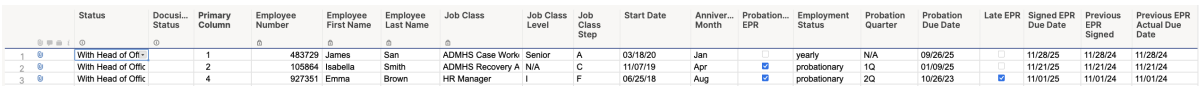
Purpose: Stores employee performance review (EPR) data, including employee details, review timelines, and multi-level approval workflow.

Key Fields (Grouped):

- **Employee Info:** Employee Number, Name, Job Class, Level, Step, Start Date
- **EPR Details:** Anniversary Month, Employment Status, Probation Info, Due Dates
- **Tracking:** Status, Late EPR, Signed EPR Dates, Previous EPR Records
- **Assignments:** Supervisor(s), HR, Head of HR, Head of Office
- **Approvals:** Supervisor, HR, and Head of Office approval statuses
- **System:** Reset Ready flag, internal reference fields

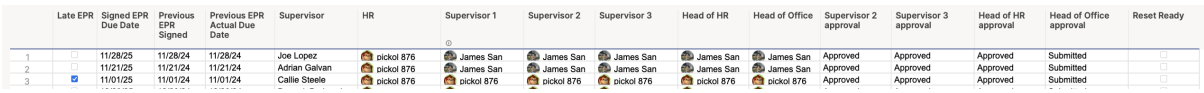
Constraints:

- Only authorized roles can update approval and status fields.



	Status	Docusi... Status	Primary Column	Employee Number	Employee First Name	Employee Last Name	Job Class	Job Class Level	Job Class Step	Start Date	Anniver... Month	Probation... EPR	Employment Status	Probation Quarter	Probation Due Date	Late EPR	Signed EPR Due Date	Previous EPR Signed	Previous EPR Actual Due Date
1	With Head of Offi		1	483729	James	San	ADMHS Case Work	Senior	A	03/18/20	Jan		yearly	N/A	09/26/25		11/28/25	11/28/24	11/28/24
2	With Head of Offic		2	105864	Isabella	Smith	ADMHS Recovery A	N/A	C	11/07/19	Apr	☑	probationary	1Q	01/09/25		11/21/25	11/21/24	11/21/24
3	With Head of Offic		4	927351	Emma	Brown	HR Manager	I	F	06/25/18	Aug	☑	probationary	2Q	10/26/23	☑	11/01/25	11/01/24	11/01/24

Figure 1: EPR 1



	Late EPR	Signed EPR Due Date	Previous EPR Signed	Previous EPR Actual Due Date	Supervisor	HR	Supervisor 1	Supervisor 2	Supervisor 3	Head of HR	Head of Office	Supervisor 2 approval	Supervisor 3 approval	Head of HR approval	Head of Office approval	Reset Ready
1		11/28/25	11/28/24	11/28/24	Joe Lopez	pickol 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Submitted	
2		11/21/25	11/21/24	11/21/24	Adrian Galvan	pickol 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Submitted	
3	☑	11/01/25	11/01/24	11/01/24	Callie Steele	pickol 876	pickol 876	pickol 876	pickol 876	pickol 876	pickol 876	Approved	Approved	Approved	Submitted	

Figure 2: EPR 2

8.3.2. EPR Tracker - Employees History Records

Purpose: Tracks the processing status of EPR records, including document handling (e.g., DocuSign), review progress, and approval workflow.

Key Fields (Grouped):

- **Processing Info:** Date Saved, Status, DocuSign Status, Primary Column
- **Employee Info:** Employee Name, Job Class, Level, Step
- **EPR Details:** Anniversary Month, Probation Info, Due Dates, Late EPR
- **Assignments:** Supervisor(s), HR, Head of HR, Head of Office
- **Approvals:** Supervisor, HR, and Head of Office approval statuses
- **Tracking:** Previous EPR dates, Employment Status, Reset Ready flag

Constraints:

- Only authorized roles can update status and approval fields.

Figure 3: EPR History 1

Figure 4: EPR History 2

8.3.3. Personnel Matters - Personnel Matters Tracker

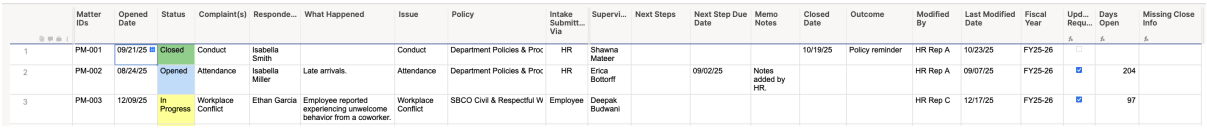
Purpose: Tracks employee relations cases, including complaints, investigation details, status, and outcomes.

Key Fields (Grouped):

- **Case Info:** Matter ID, Opened Date, Status, Days Open
- **Complaint Details:** Complaint Type, Issue, Policy, Description of Incident
- **People Involved:** Respondent, Supervisor, Intake Submitted Via
- **Case Management:** Next Steps, Due Dates, Memo Notes
- **Outcome & Closure:** Closed Date, Outcome, Missing Close Info
- **Tracking:** Modified By, Last Modified Date, Fiscal Year, Update Required

Constraints:

- Sensitive fields are restricted to authorized HR personnel.



	Matter IDs	Opened Date	Status	Complaint(s)	Responde...	What Happened	Issue	Policy	Intake Submitt... Via	Supervi...	Next Steps	Next Step Due Date	Memo Notes	Closed Date	Outcome	Modified By	Last Modified Date	Fiscal Year	Upd... Requ...	Days Open	Missing Close Info
1	PM-001	09/21/25	Closed	Conduct	Isabella Smith		Conduct	Department Policies & Proc	HR	Shawna Mateer				10/19/25	Policy reminder	HR Rep A	10/23/25	FY25-26			
2	PM-002	08/24/25	Opened	Attendance	Isabella Miller	Late arrivals.	Attendance	Department Policies & Proc	HR	Erica Bottorff		09/02/25	Notes added by HR.			HR Rep A	09/07/25	FY25-26	☑	204	
3	PM-003	12/09/25	In Progress	Workplace Conflict	Ethan Garcia	Employee reported experiencing unwelcome behavior from a coworker.	Workplace Conflict	SBCO Civil & Respectful W	Employee	Deepak Budwani						HR Rep C	12/17/25	FY25-26	☑	97	

Figure 5: Personnel Matters Data

8.3.4. Separations - Employees Separations Tracker

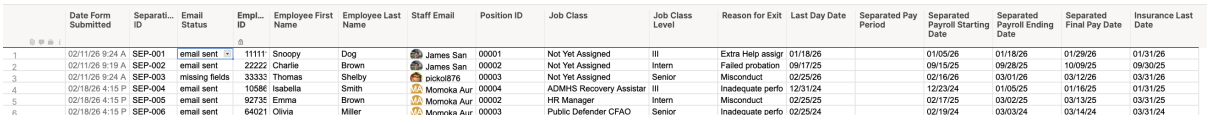
Purpose: Tracks employee separation records, including exit details, payroll timelines, and notification status.

Key Fields (Grouped):

- **Submission Info:** Date Form Submitted, Separation ID, Email Status
- **Employee Info:** Employee ID, Name, Staff Email, Position ID, Job Class, Level
- **Separation Details:** Reason for Exit, Last Day Date
- **Payroll Tracking:** Pay Period, Payroll Start/End Dates, Final Pay Date
- **Benefits:** Insurance Last Date

Constraints:

- Sensitive employee data is restricted to authorized HR personnel.



	Date Form Submitted	Separati... ID	Email Status	Empl... ID	Employee First Name	Employee Last Name	Staff Email	Position ID	Job Class	Job Class Level	Reason for Exit	Last Day Date	Separated Pay Period	Separated Payroll Starting Date	Separated Payroll Ending Date	Separated Final Pay Date	Insurance Last Date
1	02/11/26 9:24 A	SEP-001	email sent	11111	Snoopy	Dog	James San	00001	Not Yet Assigned	III	Extra Help assigr	01/18/26		01/05/26	01/18/26	01/29/26	01/31/26
2	02/11/26 9:19 A	SEP-002	email sent	22222	Charlie	Brown	James San	00002	Not Yet Assigned	Intern	Failed probation	09/17/25		09/15/25	09/28/25	10/09/25	09/30/25
3	02/11/26 9:24 A	SEP-003	missing fields	33333	Thomas	Shelby	pickol876	00003	Not Yet Assigned	Senior	Misconduct	02/25/26		02/16/26	03/01/26	03/12/26	03/31/26
4	02/18/26 4:15 P	SEP-004	email sent	10586	Isabella	Smith	Momoka Aur	00004	ADMHS Recovery Assistar	III	Inadequate perfo	12/31/24		12/23/24	01/05/25	01/16/25	01/31/25
5	02/18/26 4:15 P	SEP-005	email sent	92735	Emma	Brown	Momoka Aur	00002	HR Manager	Intern	Misconduct	02/25/25		02/17/25	03/02/25	03/13/25	03/31/25
6	02/18/26 4:15 P	SEP-006	email sent	64021	Olivia	Miller	Momoka Aur	00003	Public Defender CFAO	Senior	Inadequate perfo	02/25/24		02/19/24	03/03/24	03/14/24	03/31/24

Figure 6: Separation Data

8.3.5. Separations - Recognized Holidays

Purpose: Maintains a list of organizational holidays, including dates and rules for recurring holidays.

Key Fields (Grouped):

- **Holiday Info:** Holiday ID, Holiday Name
- **Dates:** Previous Holiday Date, Upcoming Holiday Date
- **Recurrence Rules:** Fixed Day Every Year, Fixed Day, Fixed Week of Month

Constraints:

- Recurring holiday fields must be consistently defined for accurate date calculation.

	Holiday ID	Previous Holiday Date	Upcoming Holiday Date	Holiday Name	Fixed Day Every Year?	Fixed Day	Fixed Week of Month
1	HOLIDAY-01	01/01/26	01/01/27	New Year's Day	☐	N/A	N/A
2	HOLIDAY-02	01/19/26	01/18/27	M.L.King Jr.'s Birthday	☑	Monday	Third Week of Month
3	HOLIDAY-03	02/12/26	02/12/27	Lincoln's Birthday	☐	N/A	N/A
4	HOLIDAY-04	02/16/26	02/15/27	Washington's Birthday	☑	Monday	Third Week of Month
5	HOLIDAY-05	05/26/25	05/26/26	Memorial Day	☑	Monday	Last Week of Month
6	HOLIDAY-06	06/19/25	06/19/26	Juneteenth	☐	N/A	N/A
7	HOLIDAY-07	07/04/25	07/04/26	Independence Day	☐	N/A	N/A
8	HOLIDAY-08	09/01/25	09/07/26	Labor Day	☑	Monday	First Week of Month
9	HOLIDAY-09	10/13/25	10/12/26	Columbus Day	☑	Monday	Second Week of Month
10	HOLIDAY-10	11/11/25	11/11/26	Veteran's Day	☐	N/A	N/A
11	HOLIDAY-11	11/27/25	11/26/26	Thanksgiving Day	☑	Thursday	Fourth Week of Month
12	HOLIDAY-12	11/28/25	11/27/26	Thanksgiving Friday	☑	Friday	Fourth Week of Month
13	HOLIDAY-13	12/25/25	12/25/26	Christmas Day	☐	N/A	N/A

Figure 7: Separation Holiday Data

8.3.6. Vacancies and Recruitment - Vacancies and Recruitment

Purpose: Tracks job vacancies, recruitment progress, and hiring activity.

Key Fields (Grouped):

- **Vacancy Info:** Vacancy ID, Department, Position ID, Job Class Title
- **Status Tracking:** Status (e.g., Posted, Screening, Interviewing), Label
- **Timeline:** Vacancy Start Date, Date Filled, Vacancy Days Open
- **Recruitment Activity:** Recruitment Type, Interviews Conducted
- **System:** Row Created Timestamp

Constraints:

- Limited by SharePoint list structure.
- Recruitment status fields are updated by authorized HR personnel only.

	Row Created Timestamp	Vacan... ID	Dept	PosID	JobClassTitle	Status	Label	Vacancy Start Date	Date Filled	Vacancy Days Open	Recruitment Type	Interviews Conducted
1	02/21/26 1:25 AM	VAC-002	023	22911	COMPUTER SYSTEMS S	POSTED	FUNDED	02/20/26		24	INTERNAL	4
2	02/21/26 1:25 AM	VAC-003	023	25530	DEP PUBLIC DEFENDER	SCREENING	FUNDED	02/20/26		24	INTERNAL	1
3	02/21/26 1:25 AM	VAC-004	023	7518	DEP PUBLIC DEFENDER	INTERVIEWING	FUNDED	02/20/26		24	INTERNAL	2
4	02/21/26 1:25 AM	VAC-005	023	16009	DEP PUBLIC DEFENDER	OFFER EXTEND	FUNDED	02/20/26		24	INTERNAL	3
5	02/21/26 1:25 AM	VAC-006	023	22099	DEP PUBLIC DEFENDER	OFFER ACCEPT	FUNDED	02/20/26		24	EXTERNAL	1

Figure 8: Vacancies and Recruitments Data

This single-sheet data structure provides modularity, real-time synchronization, and high traceability while maintaining compliance with SBC PDO HR data governance policies.

9. User Interface

9.1. Interface Overview

The system does not implement a custom front-end application. User interaction occurs through three existing SaaS platforms:

9.1.1. EPR

Smartsheet Interface and Notifications

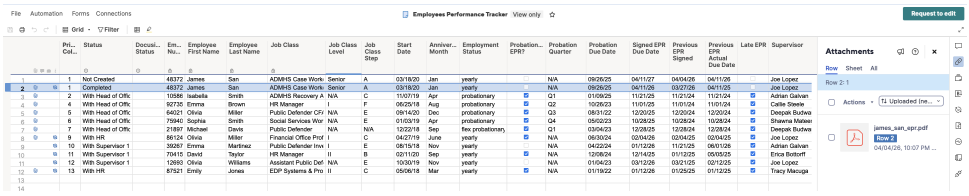


Figure 9: EPR Smartsheet page

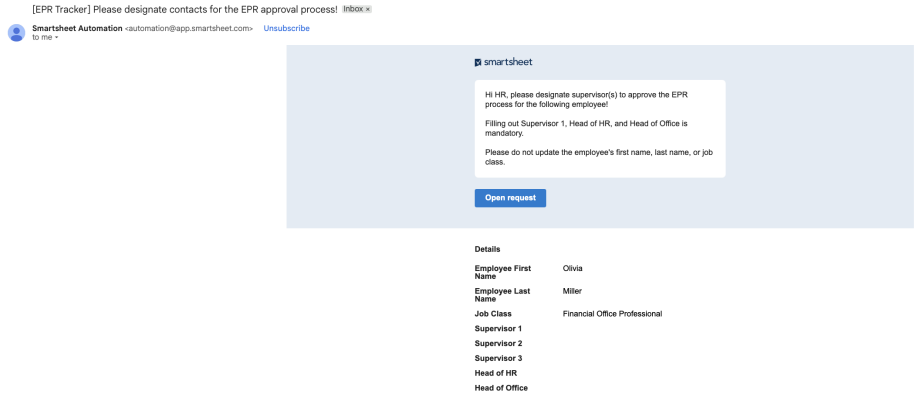


Figure 10: Automated Smartsheet emails prompting data from HR

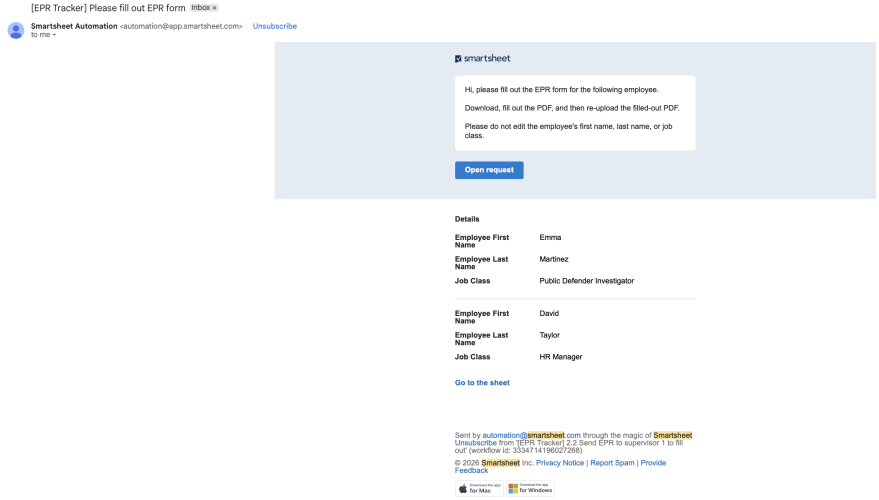


Figure 11: Smartsheet-generated email prompting user actions

Power BI Dashboard

The EPR Analytics Dashboard in Power BI provides near real-time insights into staffing and Employee Performance Review (EPR) progress. Data is refreshed nightly from Smartsheet.

Page 1 – HR Overview **Purpose:** Provide a high-level view of workforce structure, job classifications, and reporting relationships.

Key Features:

- **Summary Cards (Employees, Supervisors, Interns, Employees on Probation)** Displays key workforce metrics that dynamically update based on filters and slicers.
- **Find Employee (Search Slicer)** Allows users to search and filter the dashboard by one or more employees.
- **Employee Directory** A table showing Full Name, Job Class, Job Class Level, and Supervisor. Selecting a row filters other visuals.
- **Employees by Job Class Level** Bar chart showing distribution across levels (Intern, I, II, III, Senior).
- **Employees by Job Class** Horizontal bar chart displaying employee counts by job category.
- **Employees by Job Class and Level (Stacked Bar Chart)** Combined visualization showing employee count by job class segmented by level.

Use: Supports workforce analysis, staffing distribution insights, and organizational planning.

Page 2 – EPR Performance Review **Purpose:** Track the status and completion of Employee Performance Reviews.

Key Features:

- **Date Range Filter** Allows users to view EPRs within a selected time window.
- **KPI Cards** Displays Open EPRs, Completed EPRs, Total EPRs, and Completion Rate.
- **Open vs. Completed by Month** Line chart showing trends in review progress over time.
- **Open EPRs by Job Class** Bar chart highlighting outstanding reviews by job category.

Key Metrics:

- Completion Rate (%) = Completed EPRs / Total EPRs
- Open EPRs = Reviews currently in progress
- Overdue EPRs = Reviews past due and not completed

Use: Helps HR and supervisors monitor review progress, identify overdue evaluations, and track completion performance.

9.1.2. Personnel Matters

Smartsheet / Intake Interfaces

Figure 12: Personnel Matters Employee Intake Form

Figure 13: Personnel Matters HR Intake Form

Smartsheet / Data Tracker Interface

Key Features:

- **KPI Cards** display summary values such as Total Matters, Open Matters, Closed Matters, and Average Days to Resolution.
- **Status Breakdown** visual shows the distribution of cases by current status to help users quickly identify unresolved or in-progress matters.
- **Matters Opened Over Time** chart displays monthly or yearly intake trends for personnel matters.
- **Matters Closed Over Time** chart shows case resolution activity across time.
- **Detailed Tracker Table** displays case-level information such as Matter ID, Opened Date, Status, Complaint Type, Supervisor, and Closed Date.
- **Filter Controls** allow users to refine the dashboard by date range, fiscal year, status, complaint category, supervisor, or intake method.

Use: Helps HR monitor case progress, identify overdue or unresolved matters, support compliance reporting, and improve visibility into employee relations.

9.1.3. Vacancies and Recruitment

Smartsheet / Data Import Interface

The Vacancies and Recruitment component uses structured vacancy and recruitment data to support reporting on open positions, hiring progress, and workforce planning.

File Automation Forms Connections — Vacancies and Recruitments View only

Grid Filter

	Row Created Timestamp	Vacancy ID	Dept	PosID	JobClassTitle	Status	Label	Vacancy Start Date	Date Filled	Vacancy Days Open	Recruitment Type	Interviews Conducted
1	02/21/26 1:25 AM	VAC-002	023	22911	COMPUTER SYSTEMS SPEC I	POSTED	FUNDED	02/20/26		48	INTERNAL	4
2	02/21/26 1:25 AM	VAC-003	023	25530	DEP PUBLIC DEFENDER I	SCREENING	FUNDED	02/20/26		48	INTERNAL	1
3	02/21/26 1:25 AM	VAC-004	023	7518	DEP PUBLIC DEFENDER I	INTERVIEWING	FUNDED	02/20/26		48	INTERNAL	2
4	02/21/26 1:25 AM	VAC-005	023	16009	DEP PUBLIC DEFENDER I	OFFER EXTENDED	FUNDED	02/20/26		48	INTERNAL	3
5	02/21/26 1:25 AM	VAC-006	023	22099	DEP PUBLIC DEFENDER I	OFFER ACCEPTED	FUNDED	02/20/26		48	EXTERNAL	1
6	02/21/26 1:25 AM	VAC-007	023	25531	LEGAL OFFICE PRO I	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
7	02/21/26 1:25 AM	VAC-008	023	25550	LEGAL OFFICE PRO I	ON HOLD	UNFUNDED	02/20/26		48	EXTERNAL	2
8	02/21/26 1:25 AM	VAC-009	023	23629	LEGAL OFFICE PRO I	ON HOLD	UNFUNDED	02/20/26		48	INTERNAL	3
9	02/21/26 1:25 AM	VAC-010	023	7495	FINANCIAL OFFICE PRO SR	FILLED	FUNDED	02/20/26	02/20/26	0	INTERNAL	0
10	02/21/26 1:25 AM	VAC-011	023	7497	LEGAL OFFICE PRO SR	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
11	02/21/26 1:25 AM	VAC-012	023	22917	SOCIAL SERVICES WORKER	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
12	03/06/26 9:11 AM	VAC-017	023	12345	LEGAL OFFICE PRO SR	POSTED		03/06/26		34		
13	03/06/26 9:11 AM	VAC-018	023	11112	DEP PUBLIC DEFENDER I	POSTED		03/06/26		34		
14												

Figure 16: Recruitment & Vacancies Smartsheet Data

Power BI Dashboard

Figure 17: Power BI Vacancies Dashboard Page 1

Page 1 — Vacancies Overview **Purpose:** Provide a high-level view of current vacancy volume, active positions, and vacancy distribution across the organization.

Key Features:

- **KPI Cards** display summary values such as Total Vacancies, Active Vacancies, Filled Vacancies, and average vacancy days open.
- **Active Vacancies by Job Class** visual highlights which job categories currently have the highest staffing need.
- **Vacancy Status Breakdown** visual displays positions by workflow stage, such as Posted, Screening, Interviewing, Offer Extended, or Filled.
- **Vacancy Trend Over Time** chart shows how vacancy volume changes across months or fiscal periods.
- **Additional categorical visuals** may show vacancy distribution by funding label, department, office location, or other available organizational fields.
- **Filter controls** allow users to refine the dashboard by date range, fiscal year, job class title, department, funding label, or vacancy status.

Use: Supports staffing analysis, hiring prioritization, and workforce planning by helping HR identify where vacancies are concentrated and how long positions remain open.

Page 2 — Recruitment Overview **Purpose:** Track recruitment activity, hiring progress, and operational metrics related to filling positions.

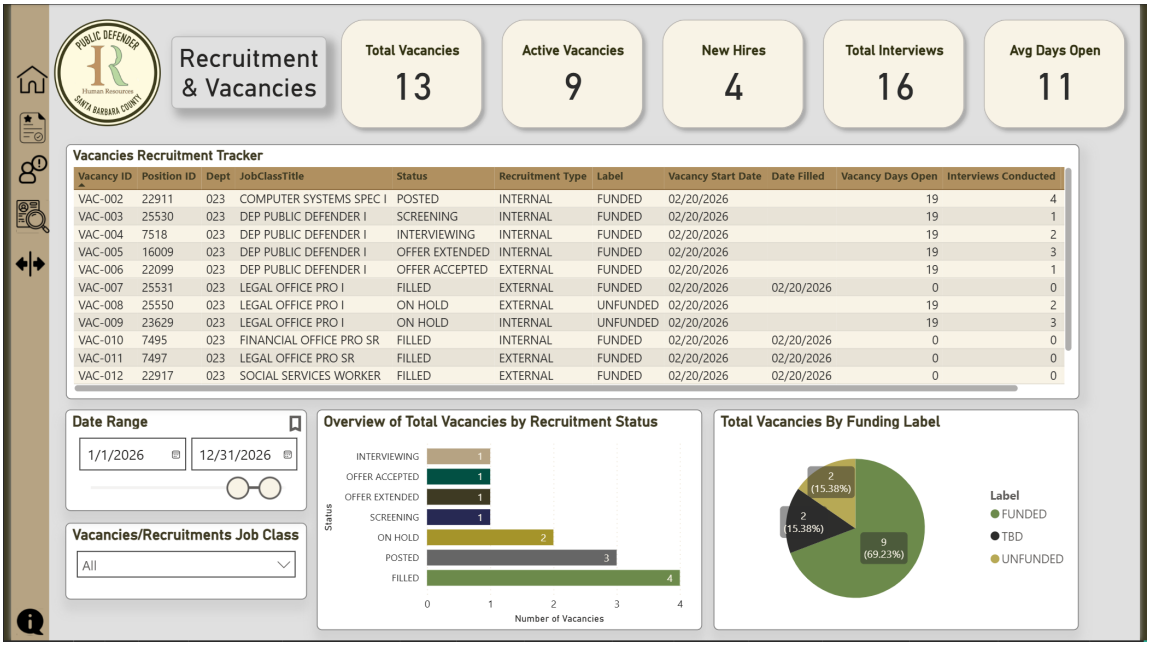


Figure 18: Power BI Vacancies Dashboard Page 2

Key Features:

- **KPI Cards** display summary values such as Recruitment Records, Interviews Conducted, New Hires, or other hiring activity metrics available in the dataset.
- **Recruitment Type Breakdown** visual shows internal versus external recruitment activity when applicable.
- **Hiring Activity Over Time** chart displays recruitment and hiring trends across selected time periods.
- **Historical Overview by Job Class** visual helps HR compare recruitment volume and outcomes across classifications.
- **Detailed Recruitment Table** provides row-level visibility into vacancy ID, job class title, recruitment type, interviews conducted, date filled, and current status.
- **Shared Filters** allow users to analyze recruitment data by job class, recruitment type, status, fiscal year, and date range.

Use: Helps HR track recruitment pipeline progress, monitor hiring outcomes, evaluate time-to-fill patterns, and support operational hiring decisions.

9.1.4. Separations

Smartsheet Form Interface

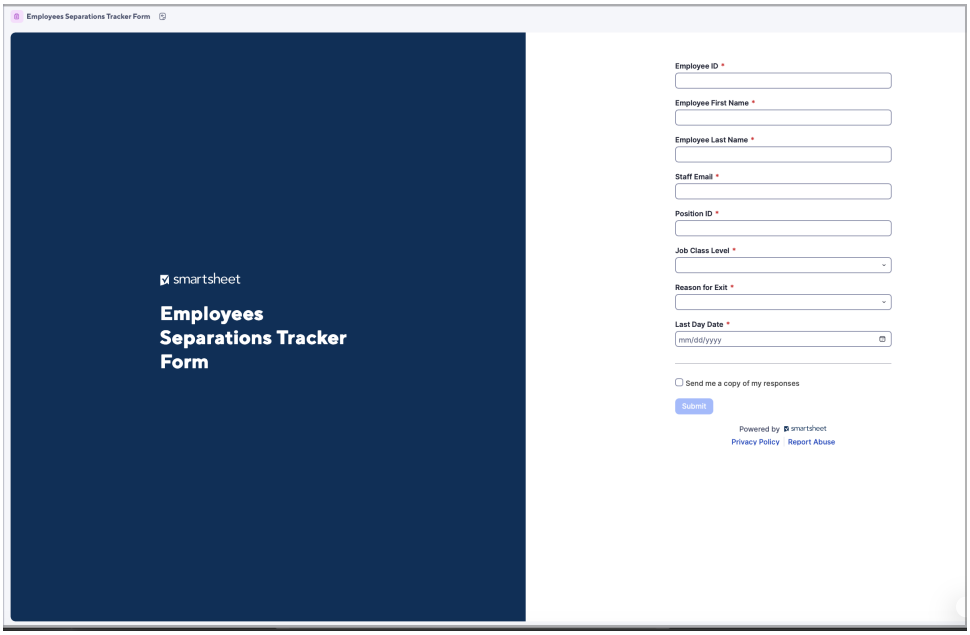


Figure 19: Separation HR Intake Form

Separations Dashboard

The Separations Dashboard in Power BI provides insights into employee separations, including total exits, reasons for leaving, and trends over time. The dashboard supports filtering by job class, fiscal year, and date range to enable flexible analysis.

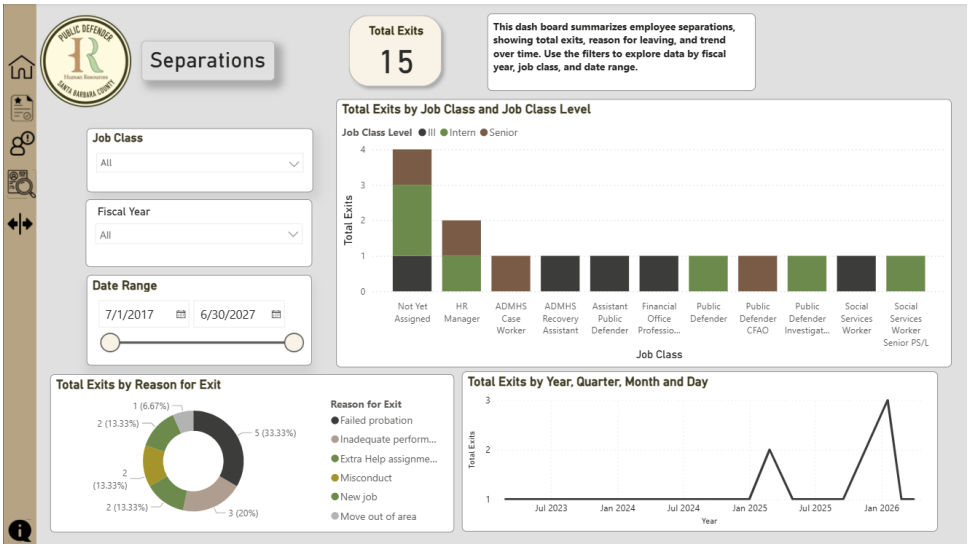


Figure 20: Separation PowerBI Dashboard

Overview **Purpose:** Provide a comprehensive view of employee separations to support workforce planning, retention analysis, and organizational decision-making.

Key Features:

- **Total Exits KPI** Displays the total number of employee separations within the selected filters.

- **Filters (Job Class, Fiscal Year, Date Range)** Allows users to refine the dataset dynamically for targeted analysis.
- **Total Exits by Job Class and Job Class Level** Stacked bar chart showing separations across job classes, segmented by level (e.g., Intern, I, Senior). *Use:* Helps identify which roles and levels experience the highest turnover.
- **Total Exits by Reason for Exit** Donut chart displaying the distribution of separation reasons (e.g., failed probation, new job, misconduct). *Use:* Supports understanding of primary drivers of employee turnover.
- **Total Exits Over Time** Line chart showing separation trends by year, quarter, month, and day. *Use:* Enables identification of patterns or spikes in employee exits over time.

Use: Supports HR in monitoring attrition, identifying turnover patterns, and informing retention strategies.

10. Requirements Validation and Verification

This section describes how key system functions of the Intelligent HR Analytics & EPR Management Dashboard are validated to ensure they meet defined requirements.

Requirement	**Component / Module**	**Testing Method**
Simple and accessible user interface for HR staff and supervisors	User Interface (Smartsheet, Power BI, Box)	Manual walkthroughs, user feedback sessions, and usability review
Ability to create, edit, and track employee performance reviews	Smartsheet Automation Rules	Functional testing by adding and updating records; verification of rule triggers
Correct routing of EPR forms for signatures	DocuSign Integration	Integration testing with sample forms; validation of signing order and completion notifications
Automatic storage of completed EPRs in Box	Box Storage Connector	File upload verification; validation of folder structure and access permissions
Display of EPR status and completion data in dashboards	Power BI Dashboard	Comparison of dashboard data with Smartsheet records; testing of filters and visuals
Daily update of Power BI datasets	Data Services Layer	Verification of scheduled refresh and dataset timestamps
Enforcement of user roles and permissions	Smartsheet / Box Permissions	Role-based access testing for HR, supervisors, and employees
Protection of data through secure access and encryption	Smartsheet, Box	Verification of HTTPS usage and compliance with County IT security policies
Support for multiple concurrent users without performance degradation	Smartsheet / Integration Layer	Simulation of concurrent updates and observation of system responsiveness
Recording of system activity for auditing purposes	Smartsheet History Log	Review of logs for accuracy of timestamps, user actions, and changes

All major modules were validated through user testing and workflow simulations to ensure the system operates correctly, securely, and efficiently.

11. Installation Guide

11.1. Repository Setup

- Clone the Git repository.
- Pull the latest changes from the repository.
- Place the `.env` file in the root folder (same level as the main project files).

11.2. Virtual Environment Setup

- Create a virtual environment:

```
python -m venv venv
```

- Activate the virtual environment:

```
source venv/bin/activate
```

11.3. Python Version Requirement

- The project requires **Python 3.11**.
- If Python 3.11 is not installed, a possible solution is:
  - Install `pyenv`.
  - Install Python 3.11 using `pyenv`.
  - This will create a `.python-version` file in the repository.
  - The repository should now use Python 3.11 automatically.
- *Note: This method may not work in all environments.*

11.4. Install Dependencies

- Install required packages:

```
pip install -r requirements.txt
```

- Most dependencies require Python 3.11.

11.5. Separations Project Setup

- Configure the email password in the `.env` file using the provided link.
- Update the sender address in:

```
layers/shared_config/config.py
```

to use the environment variable.

12. Glossary

- **EPR (Employee Performance Review):** A structured evaluation process for employee performance and development.
- **HR (Human Resources):** The department responsible for employee records, reviews, and workflow management.
- **Smartsheet:** Cloud-based platform used to manage employee data, workflow automation, and EPR tracking.
- **Box:** Cloud storage system used to archive finalized and signed EPR documents.
- **Power BI:** Data analytics tool used to visualize EPR metrics, progress, and workforce summaries.
- **Power Automate:** Automation service used to link Smartsheet, DocuSign, Box, and Power BI for workflow synchronization.
- **Dashboard:** A Power BI visual interface displaying HR analytics and EPR tracking metrics.
- **OAuth 2.0:** Authentication method used to ensure secure access between connected services.
- **KPI (Key Performance Indicator):** Quantifiable metric used to measure progress toward HR goals.
- **SRS (Software Requirements Specification):** The document that defines all functional and non-functional system requirements.
- **SDS (Software Design Specification):** This document; describes how the system is structured, integrated, and implemented.

13. References

- Software Requirements Specification – Intelligent HR Analytics & EPR Management Dashboard
- Smartsheet Product Documentation: https://help.smartsheet.com/
- Microsoft Power BI Documentation: https://learn.microsoft.com/power-bi/
- Box API and Security Overview: https://developer.box.com/
- Santa Barbara County Public Defender's Office – HR Process Guidelines (Internal Reference)
- Ascent (CSULA Senior Design Project Portal) – Team Sun-1 Documentation (2025): https://ascent.cysun.org/project/project/view/251